

Practice 6: Dijkstra's Algorithm

Deadline: Apr.15/16/17. Must be checked onsite (during labs).Online checks via email or messaging apps are not accepted.

1. Objective

This Practice consists of two questions based on **Chapter 4.4 Shortest Path in Graph** in the lecture. It is intended to help students understand the basic concepts of Dijkstra's Algorithm .

2. Task

Shortest Path by using Dijkstra's Algorithm

Dijkstra's algorithm solves the **single-source shortest path problem** on a weighted graph with **non-negative edge weights**.

Description:

Given n nodes, m edges, and the weight of each edge (weights are positive integers), find the shortest path from node 0 to every other node. If there is no path, output -1.

Input Description:

The first line contains two integers **n** and **m**, representing n nodes and m edges, separated by a space.

The next **m** lines each contain three integers: the first two integers represent the node indices, and the third integer represents the **distance** between the two nodes.

Output Description:

Output the shortest distances from node 0 to the other nodes in order, separated by a space.

Input Test case1:

```
9 14
0 1 4
1 7 11
0 7 8
1 2 8
2 8 2
8 7 7
7 6 1
8 6 6
2 3 7
```

```
2 5 4
6 5 2
3 5 14
5 4 10
3 4 9
```

Output

```
0 4 12 19 21 11 9 8 14
```

Input Test case2:

```
6 5
0 1 7
0 2 6
2 3 5
1 3 2
2 4 6
```

Output

```
0 7 6 9 12 -1
```

Input Test case3:

In the File `1.in`.

You should finish it in **1.5 second in Java and 1 second in C++**.

In this section, you can try the command:

Java

1. Mac or Linux

```
time java Main < 1.in
```

2. Windows power shell

```
Measure-Command { java Main < 1.in }
or
(Measure-Command { java Main < 1.in }).TotalMilliseconds
```

C++

1. Mac or Linux

```
time Main < 1.in
```

2. Windows power shell

```
Measure-Command { .\Main.exe < 1.in }  
or  
(Measure-Command { .\Main.exe < 1.in }).TotalMilliseconds
```

Output Test case3:

```
0 110 136 134 114 92 113 117 120 134 129 135 96 120 105 139 195 85 115 94 156  
121 89 90 214 109 79 124 128 99 173 117 33 133 97 110 95 63 114 175 139 160 124  
110 141 113 139 107 104 98 117 184 118 133 100 113 120 127 137 57 126 85 119  
132 112 146 155 121 157 141 128 131 115 115 108 151 159 136 108 120 104 145 141  
64 136 116 103 160 138 108 133 125 113 102 91 104 63 115 116 126 96 129 138 124  
124 153 137 108 112 105 160 98 193 90 147 132 117 126 141 129 93 100 140 121  
127 158 107 89 232 143 127 128 136 124 115 90 136 92 110 136 82 133 151 122 115  
117 142 105 125 104 68 131 129 138 125 62 95 107 110 124 168 101 125 251 247  
150 112 188 136 93 103 138 133 112 110 121 156 116 116 111 96 90 100 174 105  
139 78 124 102 150 138 91 144 154 158 108 109 129 92 104 107 140 107 121 129 86  
94 126 181 96 210 78 131 122 92 110 93 131 120 107 95 81 130 128 153 142 59 213  
103 110 84 112 120 218 132 119 142 129 152 112 118 97 113 119 146 119 125 73  
.....
```

3. Submission

Show the following to your tutor or SA **onsite**:

Grading:

- Submissions on time:
 - 0.5 points for first two test cases per question.
 - 0.5 points for the File `1.in` . **Other than that:** we will check whether using `PriorityQueue` . Whether you can finish it on time.
- Late submissions: 0 points.